O-I-00490B(Army-MR)
September 4, 1974
Interim Revision of
Fed. Spec. O-I-490A
April 26, 1965

INTERIM FEDERAL SPECIFICATION

INHIBITOR, CORROSION, LIQUID COOLING SYSTEM

This Interim Federal Specification was developed by the Army Materials and Mechanics Research Center, AMXMR-MS, Watertown, MA 02172, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification O-I-490A, dated April 26, 1965.

1. SCOPE

1.1 This specification covers one type and one grade of corrosion inhibitor intended for inhibiting water in automotive cooling systems (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

SS-S-535 - Sodium Borate, Decahydrate, Technical (Borax)
PPP-B-585 - Boxes, Wood, Wirebound
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated
PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Box, Fiberboard
PPP-C-96 - Cans, Metal, 28 Gage and Lighter
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, (for Carton Sealing)

FSC 6850

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this specification. Unless otherwise specified, the issue in effect on date of invitation for bids shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 1121 - Method of Test for Reserve Alkalinity of Engine Antifreeze, Antirusts, and Coolants
D 1287 - Method of Test for pH of Engine Antifreezes, Antirusts, and Coolants

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also widely distributed among technical groups and Federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Material. The raw materials used in the manufacture of the corrosion inhibitor shall be intimately mixed and processed so as to produce a uniform free-flowing product conforming to all the requirements of this specification.

3.2 Composition. The corrosion inhibitor shall consist of a blend of sodium tetraborate decahydrate and mercaptobenzothiazole mixed in the proportions necessary to conform to the percentage requirements specified in table I.

TABLE I. Chemical composition

Composition	Percent by weight		Test paragraph
	Minimum	Maximum	
Mercaptobenzothiazole	3.5	4.5	4.4.2.1
Sodium tetraborate, decahydrate	95.5	96.5	4.4.2.2

3.2.1 Sodium borate decahydrate. The sodium borate used shall be granular and conform to the requirements of SS-S-535, type II.

3.2.2 Mercaptobenzothiazole. The mercaptobenzothiazole used shall be of a technical grade and shall have a purity of not less than 92 percent.

3.3 Other chemical requirements. In addition to conforming to the requirements of 3.2, 3.2.1, 3.2.2, and table I the corrosion inhibitor shall conform to the chemical requirements specified in 3.3.1 to 3.3.4, inclusive.

3.3.1 pH value. When tested as specified in 4.4.2.3 the initial pH of the corrosion inhibitor when dissolved in ethylene glycol shall be 5.8 to 6.3 and when diluted with distilled water, the pH shall be 7.5 to 8.0.

3.3.2 Reserve alkalinity. The reserve alkalinity shall be not less than 12.0 when tested as specified in 4.4.2.4.

3.3.3 Insoluble matter. The insoluble matter shall not be greater than 0.5 percent by weight when tested as specified in 4.4.2.5.

3.3.4 Chlorides, sulfates, and carbonates. The corrosion inhibitor shall not contain chlorides, sulfates, or carbonates when tested as specified in 4.4.1.1 and 4.4.1.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

3.4 Stability. Stability shall be such that the corrosion inhibitor will conform to this specification after unheated warehouse storage in the original unopened containers for a 12 month period under any climatic conditions. The material shall show no excessive caking and shall be readily removable from the container after the storage period. Certification by the manufacturer that the material furnished will meet this requirement will be acceptable.

3.5 Workmanship. The corrosion inhibitor shall be processed to produce a clean, uniform product which complies with all the requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of all the corrosion inhibitor produced in a single batch and offered for acceptance at one time. A batch is that quantity of material subjected to some physical mixing operation to make the final product substantially uniform.

4.3 Sampling.

4.3.1 For examination of the preparation for delivery. Unless otherwise specified, a random sample of containers fully prepared for delivery shall be taken from each lot in accordance with MIL-STD-105 at inspection level II and acceptable quality level (AQL) = 2.5 percent defective for examination in accordance with 4.5.

4.3.2 For testing. For testing as required in 4.4, two containers shall be taken at random from the containers comprising the lot (see 4.2). If a sample fails to meet any of the tests required by this specification, the lot represented by the sample shall be rejected.

4.4 Tests.

4.4.1 Qualitative tests. The corrosion inhibitor shall meet the test requirements as specified in 4.4.1.1 and 4.4.1.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

4.4.1.1 Chlorides. Dissolve 0.25 g. of the corrosion inhibitor in 10 ml of distilled water. Add 10 drops of concentrated nitric acid and agitate. Add 1 ml of a 3 percent aqueous solution of silver nitrate and filter. Discard the filtrate. Wash the residue with concentrated ammonium hydroxide, recovering the filtrate. Acidify the filtrate with concentrated nitric acid. No precipitate shall form. A slight turbidity is allowed.

4.4.1.2 Sulfate and carbonates. Dissolve 0.25 g. of the corrosion inhibitor in 10 ml of warm 1:1 ethylene glycol and distilled water mixture. Add 1 ml of 5 percent aqueous solution of barium chloride. There shall be no precipitate. A slight turbidity is allowed.

4.4.2 Quantitative tests.

4.4.2.1 Mercaptobenzothiazole (MBT). Weigh 2 g. of the inhibitor to the nearest mg and extract in a Soxhlet apparatus with acetone until the drainings are colorless. Evaporate the acetone extract on a steam bath, dissolve the residue in 20 ml of hot benzene, filter and wash any insoluble matter with five 10-ml portions of warm benzene. To the combined filtrate and washings contained in a weighed 100-ml centrifuge tube add an excess of five percent filtered solution of copper oleate in benzene so that the solution remains green. Let stand overnight, centrifuge the orange-yellow precipitate, wash twice with benzene and once with petroleum ether while in the centrifuge tube. Dry to a constant weight at 140°C, cool in a desiccator and weigh rapidly. The gravimetric factor for calculating mercaptobenzothiazole weighed as the cuprous salt is 1.4556.

$$\% \text{ Mercaptobenzothiazole} = \frac{\text{Weight of precipitate X } 1.4556 \text{ X } 100}{\text{Weight of sample}}$$

4.4.2.2 Sodium tetraborate decahydrate. Dry approximately 15 gm of the corrosion inhibitor for 48 hours in a desiccator over anhydrous $CaCl_2$. Weigh approximately 10 gm to the nearest tenth of a milligram and dissolve in 300 ml of freshly boiled distilled water. Filter into a 500-milliliter volumetric flask. Wash residue with three 50-milliliter portions of boiled distilled water. Cool contents to 20°C and dilute to the mark with cooled boiled distilled water. Take a 100-milliliter aliquot for analysis. Each aliquot contains one-fifth of the sample weight.

Adjust the solution to pH 5.5 with 1N HCl, using glass electrodes and instrument described in method ASTM D 1121. Add 1 g. of mannitol and titrate to pH 8.5 with standardized 0.5 N NaOH. Add 0.5 g. of mannitol. If no change in pH is observed the titration is complete. If a change is observed, alternately titrate to pH 8.5 and add 0.5 g. portion of mannitol until no change in pH occurs on the addition of the mannitol. Calculate the percent of $Na_2B_4O_7 \cdot 10H_2O$ as follows:

$$\% \ Na_2B_4O_7 \cdot 10H_2O = \frac{(\text{ml NaOH})\ (\text{Normality of NaOH})\ (47.685)}{\text{Weight of original sample}}$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

4.4.2.3 pH value. To 1.0 g. of the corrosion inhibitor add 40 ml of ethylene glycol which has been neutralized to a pH of 7.0 ± 0.1 and 2.0 ml of distilled water. Warm and stir until the compound is dissolved and cool. The initial pH shall be measured at this point. To the solution prepared above add 80 ml of distilled water and again determine the pH. The pH determinations shall be made at 25° ± 2°C using the apparatus and procedure specified in ASTM D 1287.

4.4.2.4 Reserve alkalinity. To 0.250 ± 0.001 g. add 100 ml of distilled water. Warm and stir until dissolved. The reserve alkalinity shall be determined in accordance with ASTM D 1121. For purposes of this specification the reserve alkalinity is defined as the number of milliliters of 0.1 N hydrochloric acid required for the titration of 0.250 grams of inhibitor in 100 ml of water to a pH of 5.5.

4.4.2.5 Insoluble matter. Dissolve 10.0 g. of the corrosion inhibitor in 900 ml of a mixture of equal volumes of ethylene glycol and distilled water. Warm and stir. Filter through a suitable tared Gooch filtering crucible, rinse the container thoroughly with 1:1 glycol-water mixture, completely transferring all insoluble matter to the crucible. Wash with five 15-ml portions of the glycol-water mixture. Rinse the crucible with 10 ml of 1:1 ethanol-water mixture. Dry to constant weight at 105°C. Gain in weight is insoluble matter.

4.5 Examination of the preparation for delivery. Samples selected in accordance with 4.3.1 shall be examined for compliance with the packaging, packing, and marking requirements of section 5. In addition, each sample selected shall be examined for defects of the container and the closure, for evidence of leakage, and for unsatisfactory marking; each sample filled container shall also be weighed to determine the amount of the contents. Any container in the sample having one or more defects shall be rejected; and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105 the lot represented by the sample shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. The corrosion inhibitor shall be packaged in metal cans conforming to the requirements of PPP-C-96, type V, class 1 and shall contain 6 ounces of material as specified in table I. Unless otherwise specified, exterior surfaces of the cans shall have plan B coating, with side seam striped in accordance with the can specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

5.1.1.2 Intermediate packaging. Twenty-four unit cans of the corrosion inhibitor, packaged as specified in 5.1.1.1, shall be intermediately packaged in a snug-fitting W5c, style RSC fiberboard box conforming to PPP-B-636. In addition to box closure requirements, the box seams, corners, and manufacturer's joint shall be taped. Tape conforming to PPP-T-76 shall be used in lieu of the tape specified.

5.1.2 Level B.

5.1.2.1 Unit packaging. The corrosion inhibitor shall be packaged in a string opening, cylindrical pressed board container equipped with metal ends crimped, and shall contain 8-1/2 ounces of material as specified in table I. After crimping, the edge of the crimp shall be smooth, without excess gathering of metal or paper. The cover shall be furnished with metal ends crimped and provided with a skirt that can be slipped over the open end of the container and shall be tight enough not to fall off by its own weight when empty container is held in an inverted position.

5.1.2.2 Intermediate packaging. Twenty-four unit containers of the corrosion inhibitor, packaged as specified in 5.1.2.1 shall be intermediately packaged in a snug fitting W5c, style RSC fiberboard box conforming to PPP-B-636.

5.1.3 Level C. The corrosion inhibitor shall be packaged to afford adequate protection from the supply source to the first receiving activity for immediate use.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Four intermediate packages, packaged as specified in 5.1, shall be packed into snug-fitting exterior boxes conforming to PPP-B-621, class 2; PPP-B-601, overseas type; or PPP-B-636, V3c, style RSC. Closures and strapping of containers shall be in accordance with the applicable box specification and its appendix.

5.2.2 Level B. Four intermediate packages, packaged as specified in 5.1, shall be packed into snug-fitting exterior boxes conforming to PPP-B-636, V3c, style RSC; PPP-B-585, class 1; PPP-B-591, domestic type; or PPP-B-621, class 1. Closure and strapping of containers shall be in accordance with the applicable box specification and its appendix.

5.2.3 Level C. The corrosion inhibitor, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to first destination. Containers shall comply with the rules or regulations of carriers applicable to the mode of transportation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

5.3 Marking. In addition to the marking required by 5.3.1 or 5.3.2, each container shall be marked as follows:

"Contents" . ounces by weight (insert 6 or 8-1/2 as applicable).
"Direction for use - Use one (1) ounce of inhibitor to two (2) quarts of water in the cooling system of internal combustion engines. Do not use in liquid cooled aircraft engines."

5.3.1 Civil agencies. Marking of unit and intermediate packages and shipping containers shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required in the contract or order, marking of the interior packages and shipping containers shall be in accordance with MIL-STD-129. The nomenclature shall be "Inhibitor, Corrosion, Liquid Cooling System".

6. NOTES

6.1 Intended use. Corrosion inhibitor compound as specified herein, is intended for use in the cooling system of liquid-cooled internal combustion engines; specifically to inhibit water. It is intended for use at the optimum rate of 1 ounce of the inhibitor to 2 quarts of water. This inhibitor may be used with water-ethylene glycol antifreeze mixtures provided the glycol is inhibited with a borax base type inhibitor. This corrosion inhibitor shall not be used in the cooling system of liquid cooled aircraft engines.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Size of container required (6 or 8-1/2 ounce).
c. Quantity required, number of containers.
d. Level of packaging and packing required (see 5.1 and 5.2).

MILITARY CUSTODIAN:

Army - MR

Preparing activity:

Army - MR

Project No. 6850-A539

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

O-I-00490B(Army-MR), Inhibitor, Corrosion, Liquid Cooling System

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.